

Team Name

LEGAL VISIONARIES

Project Title

LegalEase – AI-Powered Legal Document Generator

Team Members

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PHASE 6: PROJECT TESTING

1. Introduction

The Project Testing phase verifies whether the LegalEase application works according to the requirements and design established in the previous phases.

Testing focuses on the user interface, input processing, document generation, AI integration, document preview, and document export functionality.

The testing process is performed using different input combinations to identify functional errors and verify the expected application behavior.

2. Testing Objectives

The main objectives of testing are:

1. To verify that the application starts and operates correctly.
2. To verify that users can provide the required information.
3. To verify document generation.
4. To verify AI integration.
5. To verify generated document preview.
6. To verify document editing functionality.
7. To verify TXT, DOCX, and PDF export functionality.
8. To identify and correct application errors.
9. To verify the complete application workflow.

3. Testing Approach

The following testing activities are planned:

- Functional Testing
- Input Validation Testing
- AI Generation Testing
- Integration Testing
- Document Export Testing
- User Interface Testing
- Error Handling Testing

4. Test Environment

Hardware

- Computer/Laptop
- Internet connection
- Keyboard and mouse

Software

- Python
- Streamlit
- FastAPI
- Gemini AI
- Web browser
- LegalEase application
- Required Python packages

5. Functional Test Cases

Test ID	Test Case	Input / Action	Expected Result
TC-01	Application Launch	Start LegalEase application	Application should open successfully
TC-02	Document Type	Select Agreement/Contract	Selected type should be accepted
TC-03	Party Information	Enter party details	Information should be accepted
TC-04	Terms Input	Enter terms and conditions	Terms should be accepted
TC-05	Effective Date	Enter valid date	Date should be accepted
TC-06	Generate Document	Click Generate	Legal document should be generated
TC-07	Generated Content	Review generated document	Structured content should be displayed
TC-08	Document Preview	Open generated content	Document should be visible for review

TC-09	Document Editing	Modify generated content	Changes should be reflected
TC-10	TXT Export	Download TXT	TXT document should be generated
TC-11	DOCX Export	Download DOCX	DOCX document should be generated
TC-12	PDF Export	Download PDF	PDF document should be generated

6. AI Generation Testing

AI generation is tested using different document requirements.

Test Scenario 1 – Agreement

The system is provided with:

- Document type: Agreement
- Party information
- Terms and conditions
- Effective date

Expected Result:

The system should generate a structured agreement based on the supplied information.

Test Scenario 2 – Contract

The system is provided with:

- Document type: Contract
- Party information
- Contract terms
- Effective date

Expected Result:

The system should generate a structured contract based on the supplied information.

7. Input Validation Testing

The application should be tested using:

Valid Input

All required fields are provided with appropriate information.

Expected Result:

The system should process the request and generate the document.

Missing Input

One or more required fields are left empty.

Expected Result:

The application should prevent or appropriately handle incomplete input.

Different Input Combinations

Different document types, parties, terms, and dates are tested.

Expected Result:

The system should process the different combinations appropriately.

8. Document Export Testing

The generated document is tested for different output formats.

TXT

The generated text should be available in TXT format.

DOCX

The generated document should be available as an editable Word document.

PDF

The generated document should be available as a PDF.

The exported files are opened after download to verify that the generated content is present and readable.

9. User Interface Testing

The Streamlit interface is tested for:

- Correct display of input fields.
- Correct display of buttons.
- Proper document preview.
- Readability of generated content.
- Proper download controls.
- General ease of interaction.

10. Integration Testing

Integration testing verifies communication between the major components:

Streamlit Frontend



FastAPI Backend



Gemini AI



Generated Document



Formatting



Document Export

The complete workflow is tested to ensure that the components work together.

11. Test Results

The following results are recorded based on the testing performed on the application.

Test ID	Test Description	Result
TC-01	Application launches successfully	PASS
TC-02	Agreement input accepted	PASS
TC-03	Agreement generated successfully	PASS
TC-04	Contract input accepted	PASS
TC-05	Contract generated successfully	PASS
TC-06	Generated document can be edited	PASS
TC-07	TXT export works	PASS
TC-08	DOCX export works	PASS
TC-09	PDF export works	PASS
TC-10	Exported files open correctly	PASS

Test ID	Test Description	Result
TC-11	Frontend and backend integration works	PASS
TC-12	Overall system functionality verified	PASS

12. Defect Tracking

Any problems identified during testing will be recorded using the following format:

Defect ID	Description	Severity	Status	Resolution
D-01	To be recorded if an issue is identified	-	-	-

If no defect is identified during a particular testing cycle, the result can be recorded as **No defect observed.**

13. Testing Evidence

Screenshots will be maintained as evidence of the testing process.

The evidence folder is:

D:\LegalEase\Phase-6-Project-Testing\Test-Screenshots

Recommended evidence includes:

1. Application input screen
2. Generated agreement
3. Generated contract
4. Document preview
5. TXT download
6. DOCX output
7. PDF output
8. Error/input validation screen, if applicable

14. Testing Summary

Testing is used to verify that the major LegalEase functions operate as expected.

The application is tested from user input through document generation, preview, editing, and export.

The test results will be updated as each functionality is verified.

15. Conclusion

The Project Testing phase provides a structured process for verifying the LegalEase application.

Functional, input validation, AI generation, integration, interface, and document export testing are performed to identify potential issues and confirm the expected behavior of the application.

The final testing results will be used to support the documentation and demonstration phases.